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REG NO: FA24_BSE_055

LAB Informational security

CODE:

```
def main():
    # Public values
    P = 23
    G = 9

    print("Public Prime (P):", P)
    print("Public Base (G):", G)

    # Input private keys
    a = int(input("Enter Alice's private key: "))
    b = int(input("Enter Bob's private key: "))

    # Public keys (exchange hoti hain)
    A = pow(G, a, P)
    B = pow(G, b, P)

    print("\nAlice's Public Key:", A)
    print("Bob's Public Key:", B)

    # Secret keys (final shared key)
    ka = pow(B, a, P)
    kb = pow(A, b, P)
```

```
    print("\nAlice's Secret Key:", ka)
    print("Bob's Secret Key:", kb)

    # Check match
    print("\nMatch:", ka == kb)

if __name__ == "__main__":
    main()
```

OUTPUT:

```
C:\Users\zunai\PycharmProjects\Python
Public Prime (P): 23
Public Base (G): 9
Enter Alice's private key: 127437
Enter Bob's private key: 34343
```

```
Alice's Public Key: 12
Bob's Public Key: 9

Alice's Secret Key: 12
Bob's Secret Key: 12

Match: True
```